

# ABHINAV KAUSHIK

## Full-Stack Machine Learning Engineer

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## SUMMARY

Senior AI/ML Engineer with 8+ years building production AI, ML and LLM systems across Healthcare, Telecom and AIOps. Works with LangGraph, RAG, document extraction, LLM evaluation, Python, PySpark, FastAPI, Docker and MLflow across the AI engineering lifecycle. Delivers measurable outcomes with stakeholders. Alongside this, I am building ThinkTree.AI, a non-profit AI learning platform used by NGO teachers to support students with ADHD.

## PROFESSIONAL EXPERIENCE

### Senior Machine Learning Engineer | United Health Group | Bengaluru, India Nov 2025 – Present

- Deployed a production lab-extraction agent with Claude Opus 4.7 and Amazon Textract, integrated with LIMS, handling about 1,800 lab documents.
- Engineered a production multi-agent analytics chatbot in LangGraph for AARP medical plans, using PySpark analysis and multi-turn reasoning, cutting manual analysis by 98%.
- Built automated LLM evaluation and observability pipelines with tracing and failure detection, reducing human-in-the-loop review by 43%.
- Developed a RAG-based Project Discovery Assistant in LangGraph combining semantic retrieval with LLM reasoning, reaching a 90% match rate.

### Senior Machine Learning Engineer | Viavi Solutions | Chennai, India Sep 2024 – Nov 2025

- Developed an internal RAG-based Project Discovery Assistant in LangGraph combining semantic retrieval with LLM reasoning, reaching a 90% match rate against existing organizational solutions.
- Built an event-correlation engine clustering alarms with Sentence-BERT and DBSCAN to trace the cascading effect of network alarm floods back to the root event, improving semantic alarm classification by 73.5%.
- Prototyped 15+ AI applications for AIOps and GeoPerformance clients from client requirements, with 7 moving into production or paid client work.

### Data Scientist | Viavi Solutions | Bengaluru, India Nov 2021 – Sep 2024

- Deployed a production SLA breach prediction system with LightGBM, FastAPI and MLflow, using GPT-3.5 for real-time root-cause analysis and remediation recommendations per breach, reducing downtime by 67%.
- Shipped anomaly detection and demand forecasting on 15+ KPIs across 500+ telecom network devices using Isolation Forest and Holt-Winters, enabling earlier detection and proactive network planning.
- Delivered technical POCs and consulting for 15+ clients across 3+ AIOps and ML platforms, contributing to an 11% increase in POC and mentoring revenue.

### Data Analyst | Optum | Gurugram, India Jun 2021 – Nov 2021

- Implemented a reranker for an existing recommendation system serving individual and child plan recommendations, optimizing renewal suggestions and increasing conversion by 2%.
- Developed a Python and Flask parser that automated extraction and transformation of raw data into structured datasets, feeding downstream analytics and reporting.

### Data Engineer | Nippon Data Systems | Gurugram, India Nov 2019 – May 2021

- Engineered SQL and PySpark data pipelines automating BAU medical-audit reporting on large healthcare datasets, cutting manual reporting effort by 95%.
- Wrote complex SQL queries with multi-table joins, CTEs and window functions, delivering ad-hoc reports and analyses for multiple stakeholder teams.
- Built real-time log observability for the COVID QRT team with Elasticsearch and Kibana, improving traceability efficiency by 80%.

### Associate Consultant | Saksoft | Gurugram, India Dec 2017 – Nov 2019

- Developed an XGBoost propensity model predicting personal-loan applications within 7 days of contact, enabling targeted outreach and customer prioritization.
- Built customer dashboards for financial and product insights with QlikSense, SQL and SAP HANA, increasing user engagement by 40%.

## PROJECTS

### Think Tree – ML Study Platform

- Built a non-profit AI-powered ML study platform for people with ADHD and context-switching challenges, designed to help users organize learning and maintain study context.

### RFP Multi-Agent Proposal Automation Platform

- Built a multi-agent RFP platform where an LLM supervisor orchestrates intent detection, retrieval, proposal generation, document matching and market research, with RAG over historical proposals.

## Financial Fraud Detection – Mar 2025

- Built a fraud detection and reasoning workflow on the JPMorgan Synthetic Dataset, pairing LightGBM scoring with LLM-generated explanations for interpretable financial-risk analysis.

## Restaurant Feedback Analyzer – Aspect-Based Sentiment Analysis

- Built an NLP pipeline to extract review aspects and determine sentiment at the aspect level, enabling granular insights into customer experience.

## EDUCATION

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|-------------------------------------------------------------------------------|---------------------|
| Master of Science, Machine Learning and AI   Liverpool John Moores University | Jan 2021 – Jan 2023 |
| Bachelor of Technology, Information Technology   Jamia Hamdard                | Aug 2013 – Aug 2017 |

## TECHNICAL SKILLS

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**AI / GenAI:** Generative AI, LLM, RAG, Agentic RAG, Agentic AI, LangGraph, LangChain, Claude, OpenAI, Embeddings, Cross-Encoder, Fine-Tuning, Hugging Face Transformers, NLP, Tool Orchestration, Amazon Textract

**Machine Learning:** PyTorch, Scikit-learn, XGBoost, LightGBM, Random Forest, Isolation Forest, LSTM, SBERT, DBSCAN, Time Series Forecasting, Demand Forecasting, Anomaly Detection, Statistical Modeling, Regression, Deep Learning

**Evals / Observability:** LLM Evals, LLM Evaluators, Agent Tracing, Observability, Elasticsearch, Kibana, Log Analysis

**Programming / Data:** Python, SQL, PySpark, Pandas, NumPy, FastAPI, Flask, Hive, Spark SQL, REST API, Pydantic, Unix

**MLOps / Cloud:** MLflow, Docker, Kubernetes, Airflow, DVC, Terraform, Jenkins, Bamboo, Bitbucket, GCP, Azure, Vercel

**Data Platforms:** Databricks, Databricks SQL, Apache Spark, Supabase

**Databases:** TimescaleDB, MySQL, Vector DB, pgvector, ArangoDB, Cosmos DB

**Search:** Apache Solr, Lucene Indexing, Information Retrieval

**BI:** Power BI, QlikSense, SAP HANA

## LANGUAGES

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English: Native

German: A1 – Pursuing

## ADDITIONAL INFORMATION

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- Network Intern, Airport Authority of India
  - Hadoop Intern, Ducat Technologies